

1 Nonproprietary Names

BP: Attapulgate

2 Synonyms

Actapulgate; *Attaclay*; *Attacote*; *Attagel*; attapulgis; palygorskite; palygorskite; *Pharmsorb Regular*.

3 Chemical Name and CAS Registry Number

Attapulgate [12174-11-7]

4 Empirical Formula and Molecular Weight

Attapulgate is a purified native hydrated magnesium aluminum silicate consisting of the clay mineral palygorskite, with the empirical formula $\text{Mg}(\text{Al}_{0.5-1}\text{Fe}_{0-0.5})\text{Si}_4\text{O}_{10}(\text{OH}) \cdot 4\text{H}_2\text{O}$.

5 Structural Formula

See Section 4.

6 Functional Category

Adsorbent.

7 Applications in Pharmaceutical Formulation or Technology

Attapulgate is widely used as an adsorbent in solid dosage forms. Colloidal clays (such as attapulgate) absorb considerable amounts of water to form gels and in concentrations of 2–5% w/v usually form oil-in-water emulsions. Activated attapulgate, which is attapulgate that has been carefully heated to increase its absorptive capacity, is used therapeutically as an adjunct in the management of diarrhea.

8 Description

Attapulgate occurs as a light cream colored, very fine powder. Particle size ranges depend on the grade and manufacturer.

9 Pharmacopeial Specifications

See Table I. See also Section 17.

Table I: Pharmacopeial specifications for attapulgate.

Test	BP 2009
Identification	+
Characters	+
Acidity or alkalinity (5% w/v aqueous suspension)	7.0–9.5
Adsorptive capacity	5–14%
Arsenic	≤ 8 ppm
Heavy metals	≤ 20 ppm
Acid-insoluble matter	≤ 12.5%
Water-soluble matter	≤ 0.5%
Loss on drying	≤ 17.0%
Loss on ignition	15.0–27.0%

10 Typical Properties

Acidity/alkalinity pH = 9.5 (5% w/v aqueous suspension)

Angle of repose 37.2–45.2°⁽¹⁾

Density 2.2 g/cm³

Density (tapped) 0.33 g/cm³ ⁽¹⁾

Flowability 20.9–29.6% (Carr compressibility index)⁽¹⁾

Particle size distribution

<2 μm in size for powder;

2–5 μm in size for aggregate.⁽¹⁾

11 Stability and Storage Conditions

Attapulgate can adsorb water. It should be stored in an airtight container in a cool, dry, location.

12 Incompatibilities

Attapulgate may decrease the bioavailability of some drugs such as loperamide⁽²⁾ and riboflavin.⁽³⁾ Oxidation of hydrocortisone is increased in the presence of attapulgate.⁽⁴⁾

13 Method of Manufacture

Attapulgate occurs naturally as the mineral palygorskite.

14 Safety

Attapulgate is widely used in pharmaceutical formulations and is generally regarded as an essentially nontoxic and nonirritant material. It is not absorbed following oral administration. In oral preparations, activated attapulgate up to 9 g is used in daily divided doses as an adjunct in the management of diarrhea.⁽⁵⁾

LD₅₀ (rat, IP): 0.34 g/kg

15 Handling Precautions

Observe normal precautions appropriate to the circumstances and quantity of material handled. Eye protection, gloves, and a dust mask are recommended. Attapulgate should be handled in a well-ventilated environment and dust generation should be minimized. When heated to decomposition, attapulgate emits acrid smoke and irritating fumes.

16 Regulatory Status

Included in nonparenteral medicines licensed in a number of countries worldwide including the UK and USA.

17 Related Substances

Activated attapulgate; magnesium aluminum silicate.

Activated attapulgate

Comments Activated attapulgate is a processed native magnesium aluminum silicate that has been carefully heated to increase its adsorptive capacity. Monographs for activated attapulgate are included in the BP 2009, USP 32, and other pharmacopeias. The USP 32 also includes a monograph for colloidal activated attapulgate.

18 Comments

The EINECS number for attapulgate is 302-243-0.

19 Specific References

- Viseras C, López-Galindo A. Characteristics of pharmaceutical grade phyllosilicate powders. *Pharm Dev Technol* 2000; 5(1): 47–52.
- Mboya SA, Bhargava HN. Adsorption and desorption of loperamide hydrochloride by activated attapulgis. *Am J Health Syst Pharm* 1995; 52: 2816–2818.
- Khalil SAH *et al.* Effect of attapulgate on the bioavailability of a model low dose drug (riboflavine) in humans. *Drug Dev Ind Pharm* 1987; 13: 369–382.

- 4 Cornejo J *et al.* Oxidative degradation of hydrocortisone in the presence of attapulgit. *J Pharm Sci* 1980; **69**: 945–948.
- 5 Sweetman SC, ed. *Martindale: The Complete Drug Reference*, 36th edn. London: Pharmaceutical Press, 2009; 1709.

20 General References

Anonymous. The silicates: attapulgit, kaolin, kieselguhr, magnesium trisilicate, pumice, talc. *Int J Pharm Compound* 1998; **2**(2): 162–163.

Viseras C *et al.* Characteristics of pharmaceutical grade phyllosilicate compacts. *Pharm Dev Technol* 2000; **5**(1): 53–58.

21 Author

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22 Date of Revision

10 February 2009.



Bentonite

1 Nonproprietary Names

BP: Bentonite

JP: Bentonite

PhEur: Bentonite

USP-NF: Bentonite

2 Synonyms

Albapel; bentonitum; E558; mineral soap; *Polargel*; soap clay; taylorite; *Veegum HS*; wilkinite.

3 Chemical Name and CAS Registry Number

Bentonite [1302-78-9]

4 Empirical Formula and Molecular Weight

$\text{Al}_2\text{O}_3 \cdot 4\text{SiO}_2 \cdot \text{H}_2\text{O}$ 359.16

Bentonite is a native colloidal hydrated aluminum silicate consisting mainly of montmorillonite, $\text{Al}_2\text{O}_3 \cdot 4\text{SiO}_2 \cdot \text{H}_2\text{O}$; it may also contain calcium, magnesium, and iron. The average chemical analysis is expressed as oxides, *see* Table I, in comparison with magnesium aluminum silicate.

Table I: Average chemical analysis of bentonite expressed as oxides in comparison with magnesium aluminum silicate.

	Bentonite	Magnesium aluminum silicate
Silicon dioxide	59.92%	61.1%
Aluminum oxide	19.78%	9.3%
Magnesium oxide	1.53%	13.7%
Ferric oxide	2.96%	0.9%
Calcium oxide	0.64%	2.7%
Sodium oxide	2.06%	2.9%
Potassium oxide	0.57%	0.3%

5 Structural Formula

The PhEur 6.4 describes bentonite as a natural clay containing a high proportion of montmorillonite, a native hydrated aluminum silicate in which some aluminum and silicon atoms may be replaced by other atoms such as magnesium and iron.

The USP32–NF27 describes bentonite, purified bentonite, and bentonite magma in three separate monographs. Bentonite is described as a native, colloidal, hydrated aluminum silicate; and purified bentonite is described as a colloidal montmorillonite that has been processed to remove grit and nonswellable ore compounds.

See also Section 4.

6 Functional Category

Adsorbent; stabilizing agent; suspending agent; viscosity increasing agent

7 Applications in Pharmaceutical Formulation or Technology

Bentonite is a naturally occurring hydrated aluminum silicate used primarily in the formulation of suspensions, gels, and sols, for topical pharmaceutical applications. It is also used to suspend powders in aqueous preparations and to prepare cream bases containing oil-in-water emulsifying agents.

Bentonite may also be used in oral pharmaceutical preparations, cosmetics, and food products, *see* Section 18. In oral preparations, bentonite, and other similar silicate clays, can be used to adsorb cationic drugs and so retard their release.^(1–3) Adsorbents are also used to mask the taste of certain drugs. *See* Table II.

Bentonite has been investigated as a diagnostic agent for magnetic resonance imaging.⁽⁴⁾

Therapeutically, bentonite has been investigated as an adsorbent for lithium poisoning.⁽⁵⁾

Table II: Uses of bentonite.

Use	Concentration (%)
Adsorbent (clarifying agent)	1.0–2.0
Emulsion stabilizer	1.0
Suspending agent	0.5–5.0

8 Description

Bentonite is a crystalline, claylike mineral, and is available as an odorless, pale buff, or cream to grayish-colored fine powder, which is free from grit. It consists of particles about 50–150 μm in size along with numerous particles about 1–2 μm . Microscopic examination of samples stained with alcoholic methylene blue solution reveals strongly stained blue particles. Bentonite may have a slight earthy taste.

9 Pharmacopeial Specifications

See Table III.

Table III: Pharmacopeial specifications for bentonite.

Test	JP XV	PhEur 6.4	USP32–NF27
Identification	+	+	+
Characters	+	+	—
Alkalinity	—	+	—
Microbial limit	—	$\leq 10^3$ cfu/g	+
Coarse particles	—	$\leq 0.5\%$	—
pH (2% w/v suspension)	9.0–10.5	—	9.5–10.5
Loss on drying	5.0–10.0%	$\leq 15\%$	5.0–8.0%
Arsenic	≤ 2 ppm	—	≤ 5 ppm
Lead	—	—	$\leq 0.004\%$
Heavy metals	≤ 50 ppm	≤ 50 ppm	—
Gel formation	+	—	+
Sedimentation volume	—	≤ 2 mL	—
Swelling power	≥ 20 mL	≥ 22 mL	≥ 24 mL
Fineness of powder	+	—	+

The USP32–NF27 also contains specifications for bentonite magma and purified bentonite. *See* Section 17.

10 Typical Properties

Acidity/alkalinity pH = 9.5–10.5 for a 2% w/v aqueous suspension.

Flowability No flow.

Hygroscopicity Bentonite is hygroscopic.⁽⁶⁾ *See also* Figure 1.

Moisture content 5–12%.

NIR spectra *see* Figure 2.

Solubility Practically insoluble in ethanol, fixed oils, glycerin, propan-2-ol, and water. Bentonite swells to about 12 times its original volume in water, to form viscous homogeneous suspensions, sols, or gels depending upon the concentration.